

# Experiment Report

## Logic Function Realization

Reference: GATE EE 2019 (Question 19)

Submitted By : Kalpana Kumari

Internship : Summer Internship, 2026

Organization : Indian Institute of Technology, Hyderabad

Department of Electrical Engineering

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## 1 Objective

The objective of this experiment is to analyse the given combinational logic circuit from GATE EE 2019, derive its Boolean expression, verify the corresponding truth table and identify the logic function implemented by the circuit.

## 2 Circuit Analysis

The given logic circuit consists of two NOT operations, two AND operations and one OR operation.

The first product term generated is

$$T_1 = \overline{X}Y$$

The second product term generated is

$$T_2 = X\overline{Y}$$

The final output is obtained by ORing the two product terms.

$$Z = T_1 + T_2$$

Therefore,

$$Z = \overline{X}Y + X\overline{Y}$$

This expression is the standard Boolean expression of an Exclusive-OR (XOR) gate. Hence,

$$\boxed{Z = X \oplus Y}$$

## 3 Truth Table

| $X$ | $Y$ | $\overline{X}$ | $\overline{Y}$ | $\overline{X}Y$ | $X\overline{Y}$ | $Z$ |
|-----|-----|----------------|----------------|-----------------|-----------------|-----|
| 0   | 0   | 1              | 1              | 0               | 0               | 0   |
| 0   | 1   | 1              | 0              | 1               | 0               | 1   |
| 1   | 0   | 0              | 1              | 0               | 1               | 1   |
| 1   | 1   | 0              | 0              | 0               | 0               | 0   |

Table 1: Truth table of the derived Boolean expression

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## 4 Circuit Diagram

The combinational logic circuit provided in the GATE EE 2019 question implements

$$Z = \overline{X}Y + X\overline{Y}$$

which is equivalent to an XOR gate.

19. Find the logic function implemented by the circuit given below in Fig. 3.7  
(GATE EE 2019)

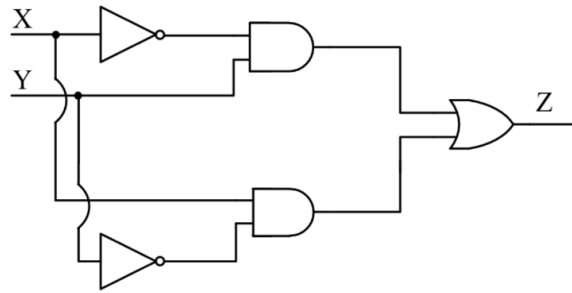


Fig. 3.7

Figure 1: Logic circuit from GATE EE 2019 Question 19

## 5 Hardware Requirements

The experiment was carried out without using any logic ICs. The following components were used:

- Breadboard
- LEDs
- 330  $\Omega$  Resistors
- Jumper Wires
- 5 V DC Supply

## 6 Experimental Procedure

1. The breadboard was powered using a regulated 5 V supply.
2. Two input lines ( $X$  and  $Y$ ) were connected manually using jumper wires.
3. LEDs were used to indicate the logic levels of the inputs and output.
4. All four input combinations were applied sequentially.
5. The observed output was compared with the theoretical truth table.

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## 7 Observation

The output LED remained OFF for input combinations (00) and (11), whereas it turned ON for (01) and (10). The observed results matched the theoretical behaviour of an XOR gate.

## 8 Result

The given combinational logic circuit implements the XOR function.

$$Z = \overline{X}Y + X\overline{Y} = X \oplus Y$$

The practical observations verified the correctness of the derived Boolean expression.

## 9 Embedded C Program

```
#include <avr/io.h>

int main(void)
{
    DDRB &= ~((1<<PB0) | (1<<PB1));
    DDRB |= (1<<PB2);

    while(1)
    {
        uint8_t x = (PINB >> PB0) & 1;
        uint8_t y = (PINB >> PB1) & 1;

        if(x ^ y)
            PORTB |= (1<<PB2);
        else
            PORTB &= ~(1<<PB2);
    }

    return 0;
}
```

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## 10 8051 Assembly Program

```
ORG 0000H

START:

LOOP:

    MOV C,P1.0
    JNC XZERO

    MOV C,P1.1
    JC BOTHHIGH

    SETB P1.2
    SJMP LOOP

BOTHHIGH:
    CLR P1.2
    SJMP LOOP

XZERO:

    MOV C,P1.1
    JC OUTPUTHIGH

    CLR P1.2
    SJMP LOOP

OUTPUTHIGH:

    SETB P1.2
    SJMP LOOP

END
```

## 11 Conclusion

The given logic circuit from GATE EE 2019 was analysed successfully. The Boolean expression was derived using Boolean algebra and verified using the truth table. Experimental verification using a breadboard, jumper wires and LEDs confirmed that the implemented logic function is an XOR gate. The same logic was also implemented through Embedded C and 8051 Assembly programs.